Group A: Quick-reference Answer Key

Q	A	Marks	Q	A	Marks	Q	A	Marks	Q	A	Marks	Q	A	Marks
1	②	1	11	①	1	21	①	2	31	②	2	41	②	4
2	②	1	12	①	1	22	④	2	32	①	2	42	①	4
3	③	1	13	④	1	23	③	2	33	①	2	43	②	4
4	④	1	14	④	1	24	①	2	34	④	4	44	③	4
5	①	1	15	②	1	25	④	2	35	④	4	45	①	4
6	③	1	16	②	1	26	④	2	36	③	4			
7	①	1	17	③	1	27	②	2	37	④	4			
8	①	1	18	①	1	28	①	2	38	②	4			
9	②	1	19	③	2	29	③	2	39	③	4			
10	③	1	20	②	2	30	①	2	40	④	4			

45 questions (18 × 1 mark, 15 × 2 marks, 12 × 4 marks), 96 marks total

Recommended Time: 2 hours 15 minutes.
Approximate Grade Boundaries (marks): A* > 77, A > 68, B > 58, C > 48, D > 39
Coverage is based on the OCR Specification.

Group A: Worked Solutions

A1. Answer: ②

The cone does not slow down during its fall, so must be accelerating downwards. The acceleration decreases from g (9.81 ms^{-2}) to zero (terminal velocity) so the magnitude of the resultant force is decreasing. (②)

A2. Answer: ②

During a phase change, the potential energy becomes less negative due to the intermolecular forces being overcome, while the kinetic energy (vibrational energy) stays the same. Since all heat energy supplied is used to break the bonds, there is no change in temperature. By definition, internal energy = kinetic energy + potential energy so this does change. (②)

A3. Answer: ③

Waves arrive constructively → 0, 2π, 4π, 6π etc phase difference.
The path difference is 20 - 16 = 4 m.
Any wavelength whose multiple includes 4 m will have a multiple of 2π phase difference. 1.0, 2.0 and 4.0 all divide into 4 m but 3.0 does not - its phase difference would be 480° (7π / 3). (③)

A4. Answer: ④

$E = hf = QV \rightarrow V$ is proportional to $f \rightarrow V$ is inversely proportional to λ. (④)

A5. Answer: ①

Higher intensity → more photons → more electrons (since one-to-one interaction) → more charge flow → more current. (①)

Work function: (minimum) energy required to remove one electron from the **highest** energy level of a surface atom.

Stopping potential: p.d. that stops the photoelectric current when the material is connected to the positive terminal of a voltage source.

Energy of photons > energy of electrons (since some is 'consumed' by the work function) → frequency of photons > frequency of electrons (as waves) → λ of photons < λ of electrons (as waves).

A6. Answer: ③

$E_k = \frac{1}{2}mv^2$ and $p = mv$. Eliminating v, we get $E_k = \frac{1}{2}\frac{p^2}{m}$, then rearranging gives $m = \frac{1}{2}\frac{p^2}{E_k} = 3$ kg. (③)

A7. Answer: ①

Since voltage is constant and $V = IR$, it follows that $I \propto \frac{1}{R}$.

Therefore, when T is low, R is high, I is low; when T is high, R is low, I is high. So the temperature scale will follow the direction of the current scale (either **A** or **B**).

At low temperatures, a fixed variation ΔT gives a very large ΔR. By considering uncertainties, it follows that $\frac{\Delta I}{I} \approx -\frac{\Delta R}{R}$. The RHS is very large, and so ΔI is also large (since the value of I is not *too* small).

Therefore, an increase in temperature from 0 ºC to 20 ºC requires a disproportionately large increase in current, meaning that the spacing between temperature marks must be larger at lower temperatures, matching ammeter **A**. (①)

A8. Answer: ①

Total mass = 15 kg → resultant acceleration = 2 ms^{-2} (right).
Taking a free-body cut between *X* and *Y*, taking *W* and *X* as the system, the force F_{XY} is $F_{XY} = m_{XY} \times a_{XY}$ = (3 + 4) × 2 = 14 N. Positive sign indicates to the right, which is away from the object, meaning tension. (①)

(Note that this is faster than working out each bar force one after the other.)

A9. Answer: ②

When the truck accelerates, a force acts on the truck (say, to the right). This force does not act on the air in the back of the truck, which therefore remains at rest (Newton's third law; inertia). So the air moves to the left making the back of the truck more dense. The balloon (helium) now floats in a density (and hence pressure) differential and so sways to the right, 'floating' above the dense air. (②)

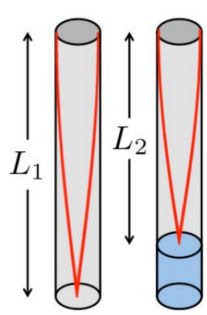
A10. Answer: ③

Chandrasekhar limit = 1.44 × Solar Mass.

Below this mass, the pressure at the core is sufficiently repelled by the electrons, but above, the electron degeneracy pressure is overcome, collapsing the star rapidly (hence unstable) and resulting in a supernova. (③)

A11. Answer: ①

Frequencies in ratio 3 : 4 → wavelengths in ratio 4 : 3 (since $v = f\lambda$).



L_1 = 20 cm → L_2 = 15 cm → depth of water = 5 cm. (①)

A12. Answer: ①

Turns ratio (step up) = 20 → current steps down by factor 20 → I_2 = 0.2 A.
Power dissipated in cables = $I^2 R$ = $0.2^2 \times 1500$ = 60 W. (①)

A13. Answer: ④

Balancing both nucleon and proton numbers,
$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{88}_{35}Br + ^{145}_{57}La + 3\ ^{1}_{0}n$
$^{145}_{57}La \rightarrow ^{145}_{56}Ba\ (+\ ^{0}_{0}e^{-} + ^{0}_{0}\bar{\nu}_e)$
Neutrons in decay product = 145 - 56 = 89. (④)

A14. Answer: ④

Diffraction slit separation = 1 / (300×10^3) = 3.333×10^{-6} m^{-1}.
Using the formula $d \sin \theta = n\lambda$, with θ at the maximum possible diffraction angle of 90° (the edges of the screen), we get n = 5.29, which rounds down to 5. Since these spots appear on both sides of the screen, there are 10 diffraction spots plus the original zeroth-order spot at θ = 0° for a total of 11 spots. (④)

A15. Answer: ②

Technetium-99m is used as it decays by double gamma emission and internal conversion from the metastable state with a half-life of 6 hours. It is synthesised in the hospital labs using a molybdenum-technetium generator. (②)

A16. Answer: ②

Intensity reflection coefficient = $\left(\frac{Z_2 - Z_1}{Z_1 + Z_2}\right)^2$, where $Z = \rho c$ is the acoustic impedance
Values of Z are then Air = 429, Water = 1.5×10^6, Muscle = 1.341×10^6,
Fat = 1.709×10^6, Bone = 6.732×10^6.

The media with the closest-matching impedances will minimise the reflection losses. Calculate to confirm: Air into Bone = 0.9997, Water into Muscle = 0.00313, Fat into Water = 0.00424, Muscle into Fat = 0.01456. (②)

A17. Answer: ③

Spring stiffness (divided by g) $k = \frac{21}{3.5 - 0.5}$ = 7 g cm^{-1}

→ self-extension = $\frac{m}{7}$ = 0.5 cm → m = 3.5 g. (③)

A18. Answer: ①

Connected in series → currents must be the same in both.
Total voltages must add to 6 V.
Only points on the graph which match this are at (2 V, 0.5 A) and (4 V, 0.5 A)
→ current = 0.5 A. (①)

A19. Answer: ③

Combine top resistors in series:	16 + 47 = 63 Ω.
Combine middle resistors:	$\left(\frac{1}{54} + \frac{1}{123 + 21}\right)^{-1}$ = 39.2727 Ω
Combine rightmost resistors in parallel:	$\left(\frac{1}{39.2727} + \frac{1}{71}\right)^{-1}$ = 25.2861 Ω
Again:	$\left(\frac{1}{25.2861 + 63} + \frac{1}{106}\right)^{-1}$ = 48.1678 Ω
Finally:	48.1678 + 17 = 65.1678 = 65.2 Ω. (③)

A20. Answer: ②

Transverse wave → particles oscillate vertically (perpendicular to direction of wave propagation).

Distance in one cycle = up, down, down, up → 4 × amplitude
= 4 × 0.04 = 0.16 m.

Time period = 1 / 10 Hz = 0.1 s → wave cycles = 20 / 0.1 = 200
Total distance = 0.16 × 200 = 32 m. (②)

A21. Answer: ①

The centripetal acceleration is $a = r\omega^2$. Consider a free-body cut of the 'triangular' section of water. It is clear that along the base, if x is the horizontal distance from the lower end, then $p = \rho x a$. Since we also have $p = \rho g h$, where h is the depth, and these pressures must be equivalent, it follows that $gh = ax \Rightarrow \frac{h}{x} = \frac{a}{g} \Rightarrow \tan\theta = \frac{r\omega^2}{g} \Rightarrow \theta = \tan^{-1}\frac{r\omega^2}{g}$. (①)

A22. Answer: ④

Diffraction occurs noticeably when the gap between two sources is close to the wavelength. $\lambda = c / f =$ (3 × 10^8) / (600 × 10^6) = 0.5 m. Distance between maxima along line at receiver $= \lambda d / a$ (from $\lambda = ax / d$) = 0.5 * 1000 / 1 = 500 m.

Since $\lambda =$ 0.5 ~ 1, diffraction effects could be significant. Since 500 m $>> \lambda$, the distances between maxima are not close. (④)

(1st option is false: frequency has no effect on diffraction)
(2nd option is false: wavelength is close to gap size)
(3rd option is false: distance between maxima is large)

A23. Answer: ③

1st option is false: 'efficiency' is with reference to the amount of energy from the fuel source which is converted into useful electrical energy – this does not refer to the amount of electrical energy which is generated or consumed.

2nd option is false: again 'efficiency' is with reference to how much of the consumed energy is converted into useful other forms – this is not relevant to the difference between generation and consumption.

4th option seems unlikely as all electronics are tested and electricity consumption is managed by select few companies.

By process of elimination, 3rd option is correct. (③)

A24. Answer: ①

Using SUVAT for linear acceleration kinematics:
$s = vt - \frac{1}{2}at^2 \rightarrow 0.4t^2 - 26t + 49 = 0 \rightarrow t = 63.06$ or 1.94 s.

Check initial speed using $v = u + at$:
If $t = 1.94$ then $u = 24.4$ ms^{-1}. If $t = 63.06$ then $u = -24.4$ ms^{-1}.
The longer time corresponds to the case where the car is initially reversing, going backwards and then eventually coming forwards to cross the bridge. This is not a reasonable way to cross the bridge so we take $t = 1.9$ s. (①)

A25. Answer: ④

Half-lives passed with X = $\frac{24}{4.8}$ = 5, Half-lives passed with Y = $\frac{24}{8}$ = 3.

Activity of X remaining = $320 \times 2^{-5} = 10$ Bq

Activity of Y remaining = $480 \times 2^{-3} = 60$ Bq

Total activity = 70 Bq. (④)

A26. Answer: ④

Towards the end of the life filament will become thinner. Considering the resistivity equation $\rho = \frac{RA}{L}$, A decreases as the filament breaks up so R must increase. Consumed power will be less ($P = V^2 / R$), so it will emit less light.

Temperature distribution will be non uniform (evaporation of tungsten and hence change in internal energy varies). At the position where temperature is maximum, the filament will break.

Black body radiation curve will become flat so the filament consumes less electrical power towards the end of the life of the bulb. (④)

A27. Answer: ②

Let c = specific heat capacity, L = specific latent heat of liquid. Let M_{cup} and C_{cup} be the mass and specific heat capacity of the container.

The first 5 g evaporates completely:
→ energy needed to heat and boil 5 g liquid = energy released
→ $(5 \times c \times (80 - 30)) + (5 \times L) = M_{cup} \times C_{cup} \times (110 - 80)$
→ $250c + 5L = 30\ M_{cup}C_{cup}$ (Eqn. 1)

The next 80 g cools the cup to 50 °C:
→ energy needed to warm 80 g liquid = energy released
→ $80 \times c \times (50 - 30) = M_{cup} \times C_{cup} \times (80 - 50)$
→ $1600c = 30\ M_{cup}C_{cup}$ (Eqn. 2)

Divide (Eqn. 1) by (Eqn. 2):
→ $(250c + 5L) / 1600c = 1$
→ $250 / 1600 + (L / c) / 320 = 1$
→ $L / c = 320 \times (1 - 250 / 1600) = 270$ (Units: K) (②)

A28. Answer: ①

Time between pulses = time period = 0.4 s.
Angular frequency = $2\pi / T = 2\pi / 0.4 = 15.7$ rad s^{-1} ~ 16 rad s^{-1}. (①)

When in the light, V across resistor = 4 V so V across LDR = 0.5 V.
When in the dark, V across resistor = 2 V so V across LDR = 2.5 V.

Current in light: $V = IR \rightarrow I = V / R = 4 / 1200 = 0.00333$ A
→ Resistance of LDR in light $= V / I = 0.5/0.00333 = 150\ \Omega$.
Current in dark: $I = 2 / 1200 = 0.001667$ A
→ Resistance of LDR in dark = 2.5 / 0.001667 = 1500 Ω.

→ Difference in resistance is 1500 - 150 = 1350 Ω.

A29. Answer: ③

Time period of light in air: λ / c = 600 nm $/ c = T$ = 2 × 10^{-15} s
Time to travel 300 nm in vacuum = 300 nm $/ c = T / 2$
Time to travel 300 nm in material = 300 nm $/ (c / 2) = T$

Therefore, after passing through the material, the light passing through one slit will be half a cycle out of phase with the other. Before, a maximum forms if path difference is $n\lambda$ or $2n\pi$ phase difference, but now since the initial phase difference is π, only π, 3π, 5π etc more phase difference is required. So the maxima form at different path differences and hence at different positions (shifted in y-direction). (③)

A30. Answer: ①

(B) is wrong: on the graph, the maximum value of Y is equal to/slightly greater than the initial value of X. This cannot be true since one nuclei of X forms one nuclei of Y and since there is still some X remaining the amount of Y must be less than its initial.

(C) is wrong: here the amount of Y appears not to decrease at all but it should since Y is also radioactive.

(D) is wrong: here the half life of Y appears to be less than/the same as that of X.

By process of elimination, the answer is A.

A31. Answer: ②

When the test tube begins to sink, the buoyant force must equal the combined weight of the test tube and the air bubble below it. Using units of cm and g,

$$\rho_w(V_{tube} + V_{air})g = (m_{tube} + m_{air})g \Rightarrow 1 \times (\frac{5}{2.5} + V_{\text{air}}) = (5 + m_{\text{air}}).$$

Since we are given that m_{air} is negligible, remove it from the equation to get $V_{\text{air}} = 3\ \text{cm}^3 \rightarrow \Delta v = 0.3\ \text{cm}^3$.

From Boyle's law (ideal gas undergoing isothermal process), since v decreased by 10%, p must have increased by 10%, so we have $p = 1 \times 10^4$ Pa. (②)

A32. Answer: ①

When the square is fully within the field ($L < x < 3L$), the induced current $I(x) = 0$ as there is no change in flux (so no e.m.f.). When moving in and out of the field, we can derive a formula for I as a function of speed v:

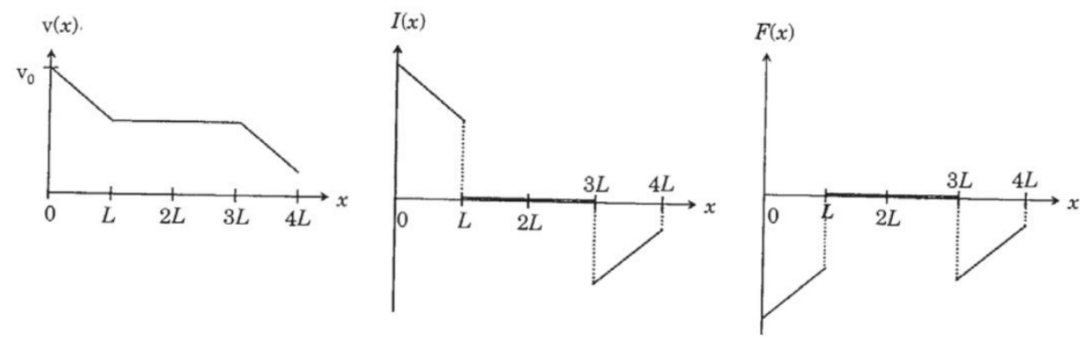
$$\varepsilon = V = IR \rightarrow I = \frac{\varepsilon}{R} = -\frac{1}{R}\frac{\Delta BA}{\Delta t} = -\frac{BL}{R}\frac{\Delta x}{\Delta t} = -\frac{BLv}{R}.$$

For $0 < x < L$, the magnetic flux is changing from zero to a positive value so there is a positive rate of change. So by Lenz's law, a force will be exerted against the motion (negative), decreasing v and hence decreasing I (by the derived formula). All graphs match.

For $L < x < 3L$, the magnetic flux density is constant so the flux linkage is constant, and so no change and hence no emf and no force (constant velocity). All graphs match this.

For $3L < x < 4L$, the magnetic flux changes from a positive value to zero so there is a negative rate of change. A force will be exerted to the left (negative) so velocity will decrease again. This time however, the rate of change of area (and hence flux linkage) means the current is induced in the opposite direction with decreasing magnitude. $I(x)$ and $F(x)$ are both different here.

The correct graphs are: (①)



A33. Answer: ①

1: The p-n junction acts as an LED with a forward bias current flowing only when the direction of flow is p → n. From the geometry of the solenoid this requires an anti-clockwise current (looking inwards from **b** towards **c**). From the right-hand screw rule, this implies the solenoid induced magnetic field is from **c** towards **b** (to the left). This implies that the solenoidal field is 'North' at **b** and 'South' at **c**, which is setup to repel the permanent magnetic field. Since the magnet enters at **c** to cause light emission, the pole closest to **c** must be the South pole, i.e. X = North pole.

2: Repeating the above analysis for a magnet with its North pole entering at **b**, we find that the current direction is the same, so the LED is still in forward-bias mode, causing photons to be emitted.

3: There are energy losses due to the light and resistive heating in the p-n junction and solenoid, so the mechanical energies will not be the same.

→ **1** and **2** only. (①)

A34. Answer: ④

Let A be the cross-sectional area of one wire. Combining in parallel,

$$R = \left(\frac{1}{R_{Cu}} + \frac{6}{R_{Al}}\right)^{-1} = \left(\frac{A}{2\rho L} + \frac{6A}{3\rho L}\right)^{-1} = \frac{2\rho L}{5A}.$$

Considering the mass of the cable,

$$M = (3d)(AL) + 6d(AL) = 9dAL \Rightarrow A = \frac{M}{9dL}$$

Eliminating A, we get $R = \frac{2\rho L \times 9dL}{5M} = \frac{18\rho dL^2}{5M}$. (④)

A35. Answer: ④

Colours with wavelength closest to diffraction spacing are diffracted through a larger angle. Diffraction spacing = 1 / (300 * 1000) = 3.3×10^{-6} m = 3300 nm.
So longer wavelengths of light (red) are closest to this (800 nm).
So red is diffracted more while violet is diffracted less.
On the graph, red has higher intensity than violet. (④)

A36. Answer: ③

Magnetic flux linkage $= BA$ cos θ where θ is the angle between the normal vector to the plane of the loop and the magnetic field. Since $\theta = \omega t$, $\Phi = BA$ cos ωt (where A is area in general).

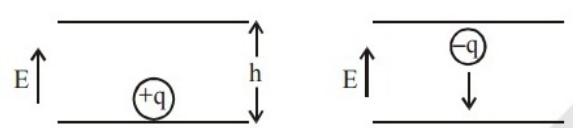
emf = - dΦ / dt $= BA\omega$ sin ωt
emf induced in smaller loop $= BA\omega$ sin ωt
emf induced in larger loop = 2 $BA\omega$ sin ωt

Notice that the emf induced in the two loops have opposite polarity (direction) because of the given orientation of the wires. So net emf = | larger - smaller | $= BA\omega$ sin ωt → 3rd option is true (they are equal) → 1st and 4th options are false (area component with flux linkage is A cos ωt but emf is proportional to sin ωt) (③)

When loop is perpendicular to paper → parallel to magnetic field → minimum (zero) flux linkage → maximum rate of change → 2nd option is false.

A37. Answer: ④

What happens to the balls next? After hitting the top plate, the balls will get negatively charged and will now get attracted to the bottom plate which is positively charged. The motion of the balls will be periodic but **not** SHM since the force in the electric fields is constant (independent of position). (④)



Variation of I_{avg}? If Q is charge on balls, then $Q \propto V_0$ (since the balls act as spherical shell capacitors and $Q = VC$). Let t be the time to rise to the top. From SUVAT and **E**-field equations,

$$h = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{F}{m}\right)t^2 = \frac{1}{2}\left(\frac{QV_0}{mh}\right)t^2 \Rightarrow t = \sqrt{\frac{2mh^2}{QV_0}}$$

Since $I_{avg} = \frac{Q}{t}$, we have $I_{avg} \propto V_0 C \times \sqrt{QV_0} \propto V_0 \times \sqrt{V_0^2} \propto V_0^2$. (④)

A38. Answer: ②

Using units of μF, V and μC for convenience,
Charge stored initially on $C_3 = Q = VC = 8$ μC.
So 3 μC have been lost to C_1 and C_2.
Capacitance of dielectric capacitor $= \varepsilon_r A / d$ but $C_2 = 1 = A / d$
→ Capacitance of $C_1 = \varepsilon_r$ μF.
Effective capacitance of the left branch $= (1 / 1 + 1 / \varepsilon_r)^{-1} = \varepsilon_r / (1 + \varepsilon_r)$.

Consider the ratio of capacitances in the left and right branches. From $Q = VC$, since the p.d. across both branches is the same (parallel), we have

Left: $Q = 3$, Right: $Q = 5$ → Capacitance of left = (3 / 5) × Capacitance of right
→ $\varepsilon_r / (1 + \varepsilon_r) = 3 / 5 \times 1$
→ $5\varepsilon_r = 3(1 + \varepsilon_r)$
→ $2\varepsilon_r = 3$
→ $\varepsilon_r = 3 / 2 = 1.50$. (②)

A39. Answer: ③

Initial activity = 2.4×10^5 Bq. Half-lives passed = 11.5 hrs / (8 × 24) = 0.06.
Activity after 11.5 hrs = $2.4 \times 10^5 \times (½)^{0.06} = 2.302 \times 10^5$ Bq.
Considering concentration ratios, $115 / (2.302 \times 10^5) = 2.5 / V$
→ $V = 5004$ ml = 5.0 litres. (③)

A40. Answer: ④

Let the volumetric flow rate be $Q = Av = 3$ m^3 s^{-1}. Using conservation of mass,
$\rho Q = \rho A_X v_X = \rho A_Y v_Y$, so $v_Y = 12$ ms^{-1}.

Using conservation of momentum (steady-flow momentum equation):
$F = \rho A_Y v_Y^{\ 2} - \rho A_X v_X^{\ 2} = 16.8$ kN. Fluid is accelerating → force acts to the right.
(Alternatively: force = mass flow rate × change in velocity)

Power = Rate of loss of mechanical energy
$= Q\,\Delta p = A_X v_X \times \frac{1}{2}\rho(v_Y^{\ 2} - v_X^{\ 2}) = 142.8$ kW. (④)

A41. Answer: ②

In 3 hours, the device transfers ($E = Pt$) = 3000 × 3 × 60 × 60 = 32.4 MJ of energy to the water circulation.

The energy required to heat the water from 10 to 30 °C (E = mc∆T):
= 120 × 4200 × 20 = 10080 J = 10.08 MJ.

So the cooler must remove (32.4 - 10.08) = 22.32 MJ:
Power = 22.32 * 10^6 / (3 × 60 × 60) = 2067 = 2100 W = 2.1 kW (2 s.f.) (②)

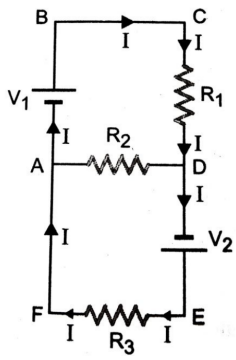
A42. Answer: ①

Consider the density of the gas in the bubble. $\rho_{gas} = m / V \rightarrow \rho_{gas} \propto 1 / V$ (since mass of gas, m, remains constant). Since temperature is constant (isothermal process), as pressure is also inversely proportional to volume (Boyle's law) so $p_{bubble} \propto \rho_{gas} \propto 1 / V$. Since $V \propto r^3$ (radius of bubble),
we now have $p_{bubble} \propto 1 / r^3$.

At any depth, pressure exerted by bubble = hydrostatic pressure + atm pressure.
At the bottom: $k / R^3 = h\rho g + P$
At a depth x: $k / r^3 = x\rho g + P$
Dividing the first equation by the second, $(k / R^3) / (k / r^3) = [h\rho g + P] / [x\rho g + P]$
$\rightarrow r^3 / R^3 = [h\rho g + P] / [x\rho g + P]$
$\rightarrow r^3 = R^3 [h\rho g + P] / [x\rho g + P]$
$\rightarrow r = R ((h\rho g + P) / (x\rho g + P))^{1/3}$. (①)

A43. Answer: ②

Let $I_2 = 0$. Assume the current will flow clockwise (if not, I will be negative).



Using KVL in the top loop, $V_1 = IR_1 \rightarrow I = \frac{V_1}{R_1}$ (Eqn. 1)

Using KVL in the whole circuit loop, $V_1 + V_2 - IR_1 - IR_3 = 0$

$\rightarrow V_1 + V_2 = I(R_1 + R_3) \rightarrow I = \frac{V_1 + V_2}{R_1 + R_3}$. (Eqn. 2)

Equating (Eqn. 1) and (Eqn. 2), we get $\frac{V_1}{R_1} = \frac{V_1 + V_2}{R_1 + R_3} \Rightarrow V_1R_3 = V_2R_1$.

Now check the given cases.

Let $V_1 = V_2 \rightarrow R_1 = R_3$ means no current (options 1 and 3)

Let $V_1 = 2V_2 \rightarrow R_1 = 2R_3$ means no current

Let $2V_1 = V_2 \rightarrow 2R_1 = R_3$ means no current (last option)

The second option does not match, so it must produce a current. (R_2 can be anything). (②)

A44. Answer: ③

Let the point at distance x be P.

Distance from S_1 to P and from S_2 to P = $\sqrt{x^2 + d^2}$.

Time for ray from S_1 = $\frac{\sqrt{x^2+d^2}}{c}$, Time for ray from S_2 $\frac{\sqrt{x^2+d^2}}{3c/4}$ [since $n = 4/3$]

So 'distance' of the second ray if it were travelling in air in the same timeframe would be $\frac{4}{3}\sqrt{x^2 + d^2}$. [we use this distance since we will combine the distances as if they were 'number of cycles']

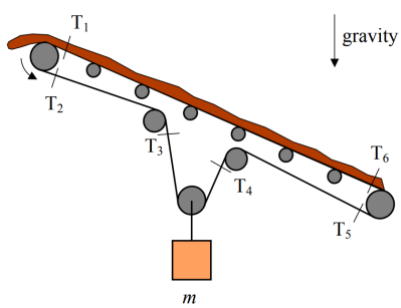
For a maximum, the path difference must be a multiple of the wavelength:

$\rightarrow \frac{4}{3}\sqrt{x^2 + d^2} - \sqrt{x^2 + d^2} = m\lambda \rightarrow \sqrt{x^2 + d^2} = 3m\lambda \rightarrow x^2 + d^2 = 9m^2\lambda^2$

$\rightarrow x^2 = 9m^2\lambda^2 - d^2 \rightarrow p^2 = 9 \rightarrow p = 3$. (③)

A45. Answer: ①

Due to the weight of the belt and the aggregate, the tension varies throughout. Consider six points of interest along the belt, at which the tensions are $T_1 \ldots T_6$:



Consider the moments acting on the top roller. Let r be the roller radius.

$T_1 r - T_2 r = M \rightarrow T_1 > T_2$.

In the straight section of the belt, the tension T_1 must support T_6 and the aggregate + belt weight:

$T_1 \sin\theta = \rho V g + M_s g + T_6 \sin\theta$ (M_s: straight belt mass, V: aggregate volume)

Considering the other two hanging sections similarly, $T_2 > T_3$, $T_4 > T_5$, $T_3 > T_4$ and $T_5 < T_6$ (due to the frictional torque).

Therefore the total ordering is

$T_1 > T_2 > T_3 > T_4 > T_6 > T_5 \rightarrow$ Maximum is at **A** (roughly), Minimum is at **C**. (①)